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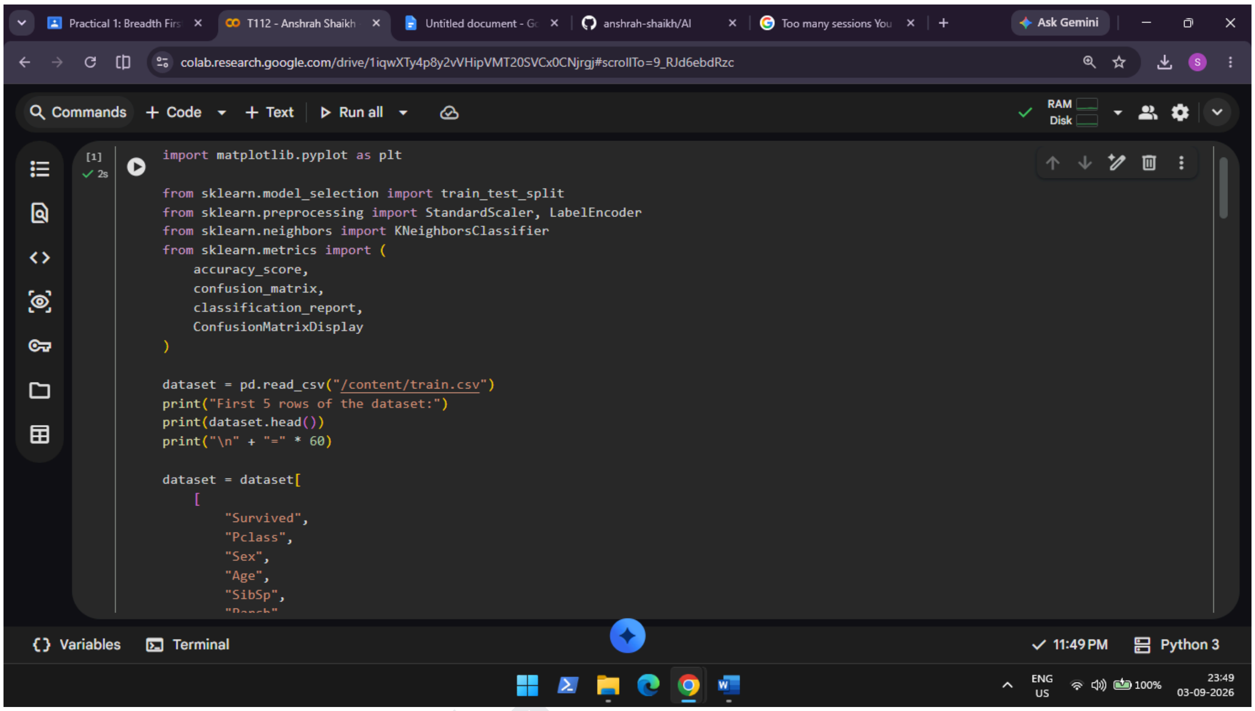
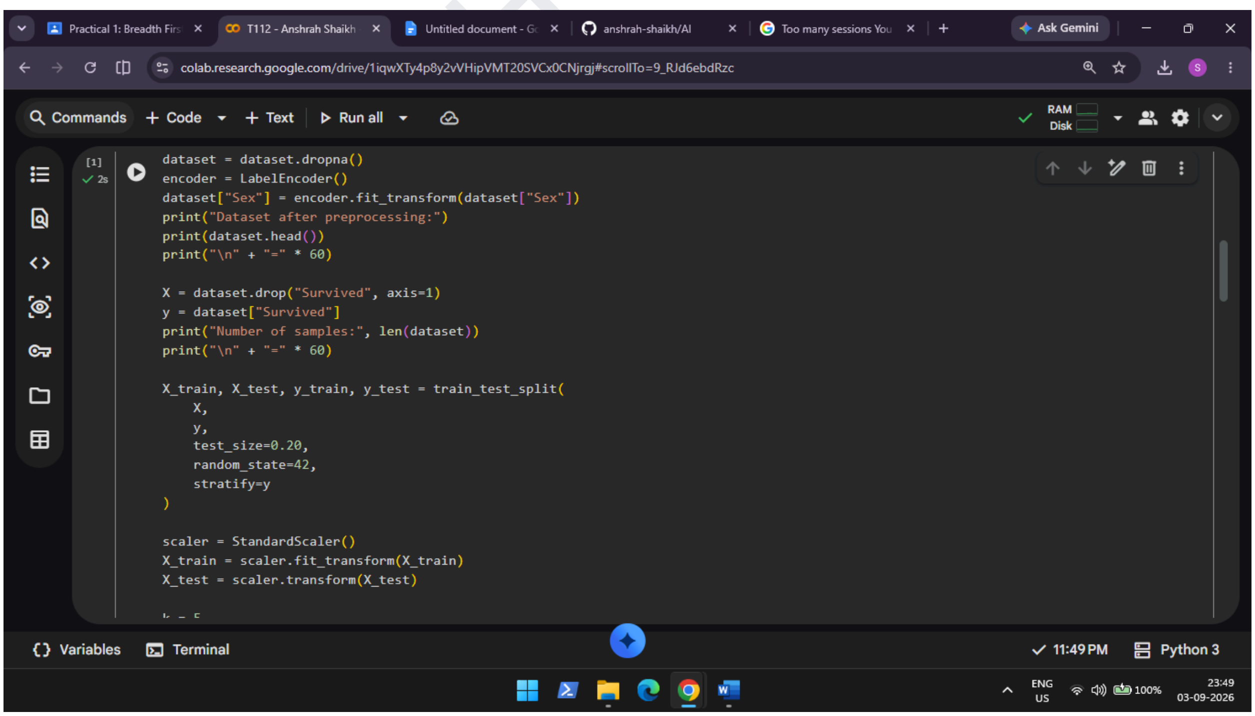
AIM: K-Nearest Neighbors (K-NN)

- Implement the K-NN algorithm for classification or regression.
- Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
- Evaluate the accuracy or error of the predictions and analyze the results.

INPUT:





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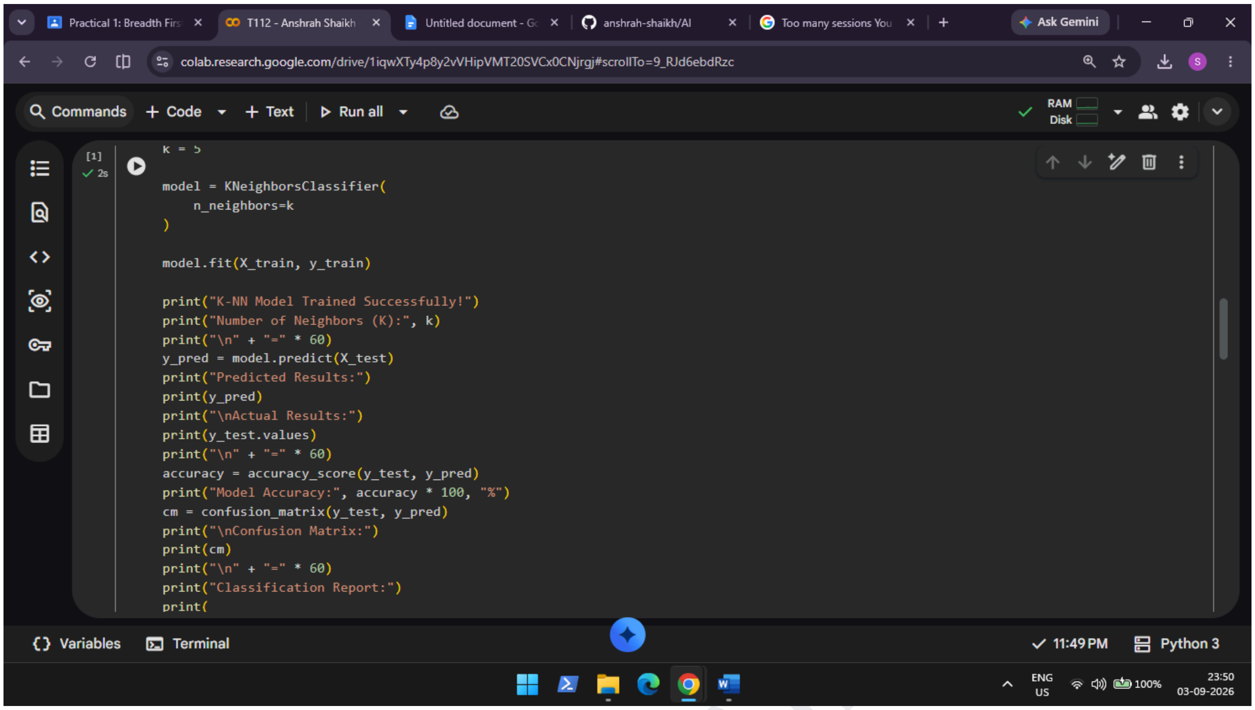
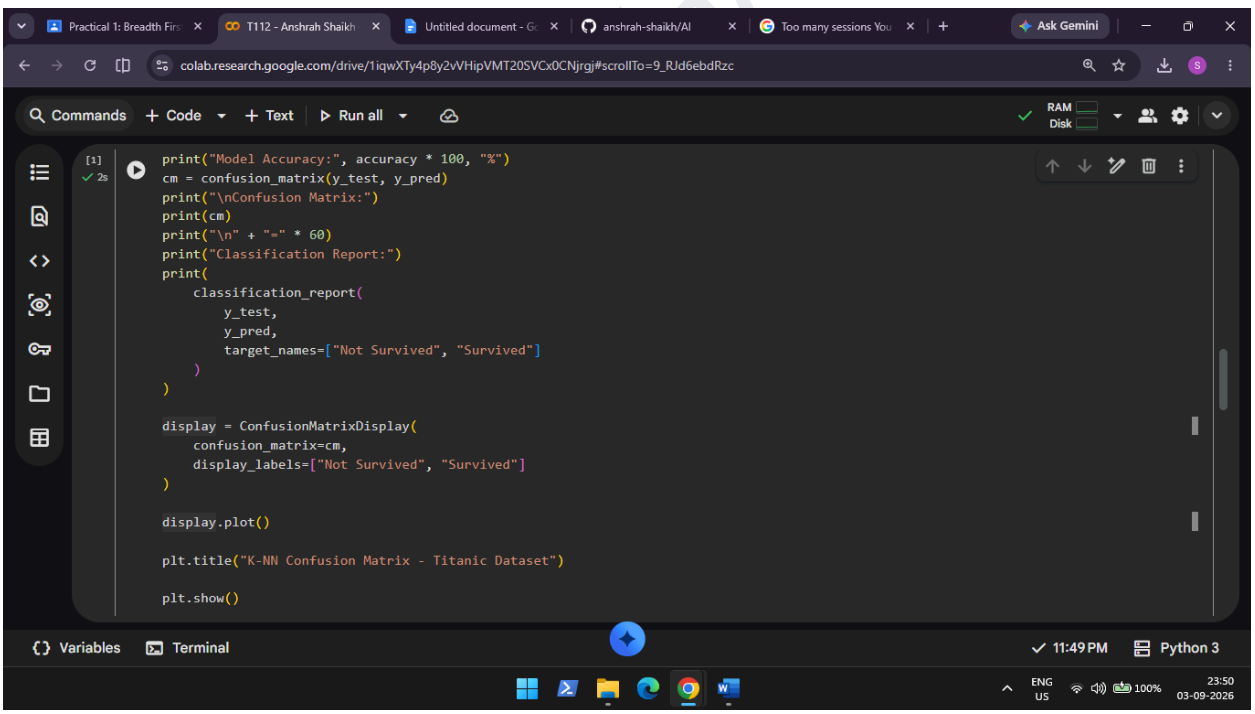


```python
k = 5

model = KNeighborsClassifier(
    n_neighbors=k
)

model.fit(X_train, y_train)

print("K-NN Model Trained Successfully!")
print("Number of Neighbors (K):", k)
print("\n" + "=" * 60)
y_pred = model.predict(X_test)
print("Predicted Results:")
print(y_pred)
print("\nActual Results:")
print(y_test.values)
print("\n" + "=" * 60)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 60)
print("Classification Report:")
print(
```



```python
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 60)
print("Classification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Not Survived", "Survived"]
    )
)

display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Not Survived", "Survived"]
)

display.plot()

plt.title("K-NN Confusion Matrix - Titanic Dataset")

plt.show()
```

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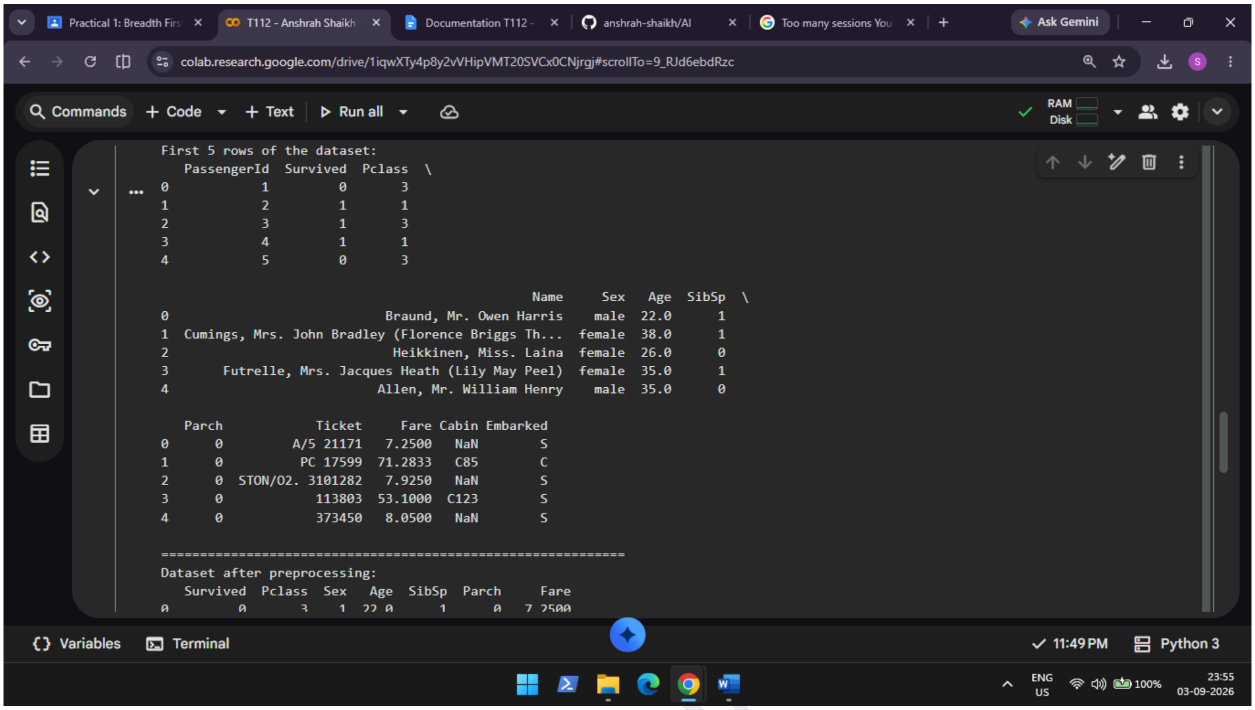
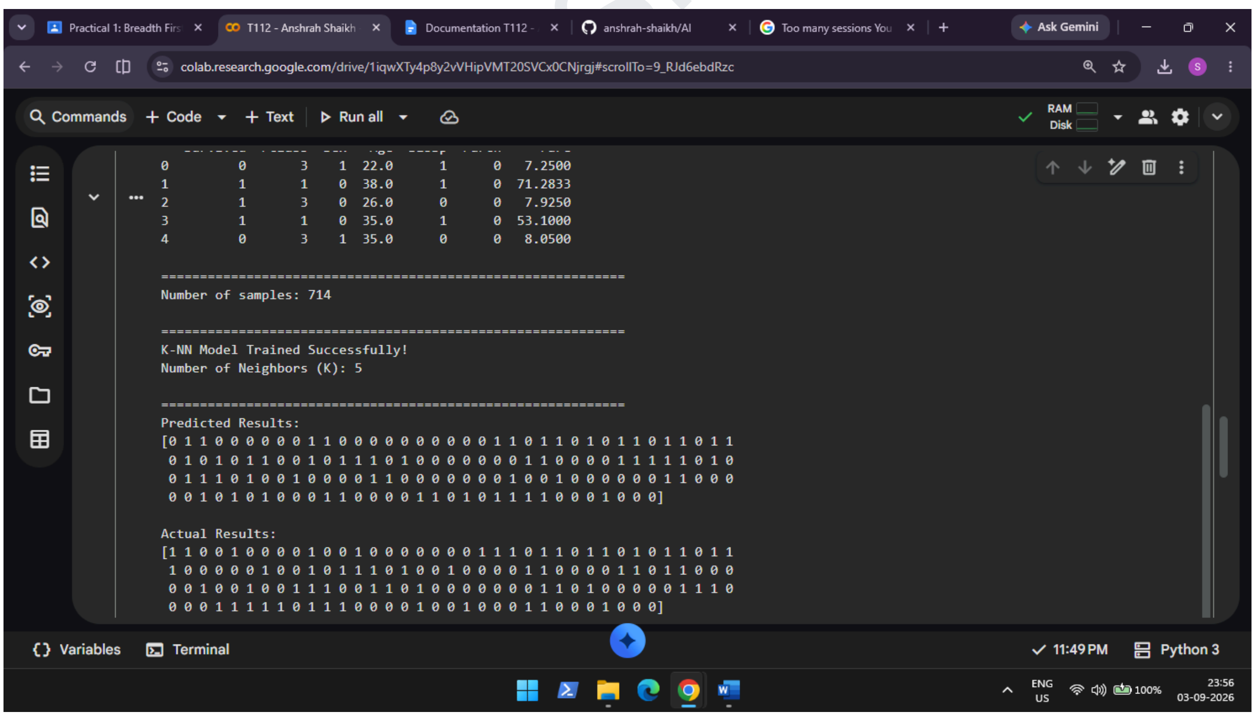
OUTPUT:





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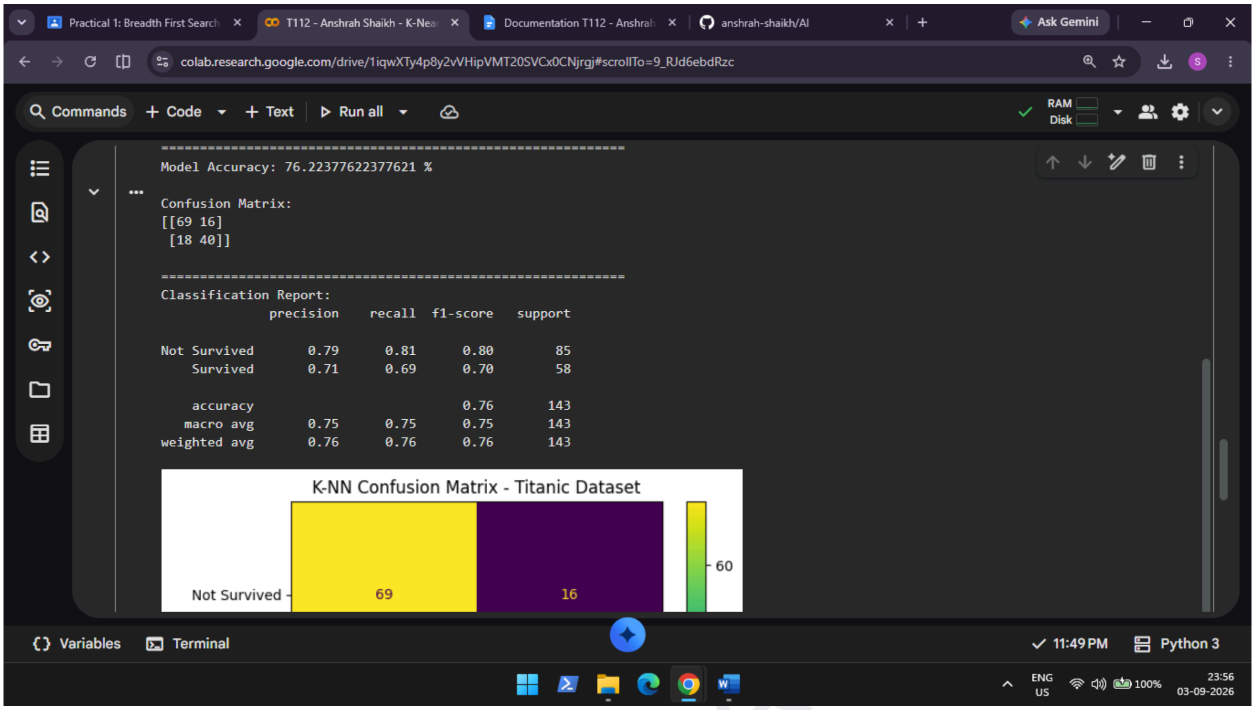

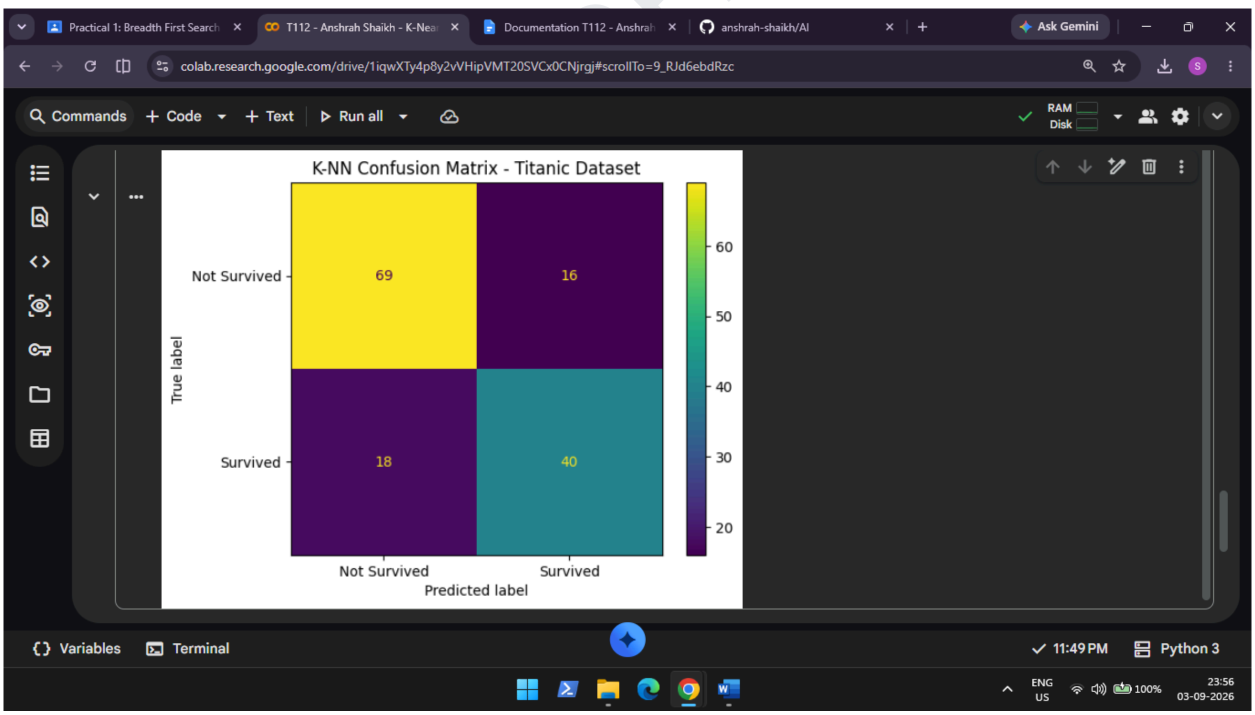

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